

46. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on

- (A) a line not passing through the origin
- (B) $|z| = \sqrt{2}$
- (C) the x -axis
- (D) the y -axis

Answer ☐ (A) ☐ (B) ☐ (C) ☒ (D)

47. Let E^c denote the complement of an event E . Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P(E^c \cap F^c | G)$ equals

- (A) $P(E^c) + P(F^c)$
- (B) $P(E^c) - P(F^c)$
- (C) $P(E^c) - P(F)$
- (D) $P(E) - P(F^c)$

Answer ☐ (A) ☐ (B) ☒ (C) ☐ (D)

48. $\frac{d^2x}{dy^2}$ equals

- (A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$
- (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$
- (C) $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$
- (D) $-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$

Answer ☐ (A) ☐ (B) ☐ (C) ☒ (D)

49. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with
- (A) variable radii and a fixed centre at $(0, 1)$
 - (B) variable radii and a fixed centre at $(0, -1)$
 - (C) fixed radius 1 and variable centres along the x -axis
 - (D) fixed radius 1 and variable centres along the y -axis

Answer ☐ ☐ ☒ ☐

(A) (B) (C) (D)

50. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct?
- (A) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$
 - (B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$
 - (C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$
 - (D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

Answer ☐ ☒ ☐ ☐

(A) (B) (C) (D)

51. Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2CD$. Let AD be perpendicular to AB and CD . If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is
- (A) 3 (B) 2 (C) $\frac{3}{2}$ (D) 1

Answer ☐ ☒ ☐ ☐

(A) (B) (C) (D)

52. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals

- (A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$ (B) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}} + K$
 (C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$ (D) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}} + K$

Answer ☒ ☐ ☐ ☐
 (A) (B) (C) (D)

53. The letters of the word **COCHIN** are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word **COCHIN** is
- (A) 360 (B) 192 (C) 96 (D) 48

Answer ☐ ☐ ☒ ☐
 (A) (B) (C) (D)

54. Consider the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

STATEMENT-1 : The parametric equations of the line of intersection of the given planes are $x = 3 + 14t$, $y = 1 + 2t$, $z = 15t$.

because

STATEMENT-2 : The vector $14\hat{i} + 2\hat{j} + 15\hat{k}$ is parallel to the line of intersection of given planes.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

Answer ☐ ☐ ☐ ☒
 (A) (B) (C) (D)

55. STATEMENT-1 : The curve $y = \frac{-x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$.

because

STATEMENT-2 : A parabola is symmetric about its axis.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
- (C) Statement-1 is True, Statement-2 is False
- (D) Statement-1 is False, Statement-2 is True

Answer

- ☒ ☐ ☐ ☐
- (A) (B) (C) (D)

56. Let $f(x) = 2 + \cos x$ for all real x .

STATEMENT-1 : For each real t , there exists a point c in $[t, t + \pi]$ such that $f'(c) = 0$.

because

STATEMENT-2 : $f(t) = f(t + 2\pi)$ for each real t .

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
- (C) Statement-1 is True, Statement-2 is False
- (D) Statement-1 is False, Statement-2 is True

Answer

- ☐ ☒ ☐ ☐
- (A) (B) (C) (D)

57. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q , respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R .

STATEMENT-1 : The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$.

because

STATEMENT-2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- (C) Statement-1 is True, Statement-2 is False
- (D) Statement-1 is False, Statement-2 is True

Answer ☐ ☐ ☒ ☐

(A) (B) (C) (D)

58. Which one of the following statements is correct?
- (A) $G_1 > G_2 > G_3 > \dots$
- (B) $G_1 < G_2 < G_3 < \dots$
- (C) $G_1 = G_2 = G_3 = \dots$
- (D) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

Answer ☐ ☐ ☒ ☐

(A) (B) (C) (D)

59. Which one of the following statements is correct?
- (A) $A_1 > A_2 > A_3 > \dots$
- (B) $A_1 < A_2 < A_3 < \dots$
- (C) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$
- (D) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

Answer ☒ ☐ ☐ ☐

(A) (B) (C) (D)

60. Which one of the following statements is correct?

(A) $H_1 > H_2 > H_3 > \dots$

(B) $H_1 < H_2 < H_3 < \dots$

(C) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$

(D) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$

M61-63: Paragraph for Question Nos. 61 to 63

If a continuous function f defined on the real line $\mathbf{R}$, assumes positive and negative values in $\mathbf{R}$ then the equation $f(x) = 0$ has a root in $\mathbf{R}$. For example, if it is known that a continuous function f on $\mathbf{R}$ is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in $\mathbf{R}$.

Consider $f(x) = ke^x - x$ for all real x where k is a real constant.

Answer ☐ ☒ ☐ ☐
(A) (B) (C) (D)

61. The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at

(A) no point

(B) one point

(C) two points

(D) more than two points

Answer ☐ ☒ ☐ ☐
(A) (B) (C) (D)

62. The positive value of k for which $ke^x - x = 0$ has only one root is

(A) $\frac{1}{e}$

(B) 1

(C) e

(D) $\log_e 2$

Answer ☒ ☐ ☐ ☐
(A) (B) (C) (D)

63. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is

(A) $\left(0, \frac{1}{e}\right)$

(B) $\left(\frac{1}{e}, 1\right)$

(C) $\left(\frac{1}{e}, \infty\right)$

(D) $(0, 1)$

Answer ☒ ☐ ☐ ☐
(A) (B) (C) (D)

64. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$.

Match the expressions/statements in **Column I** with expressions/statements in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

Column II

(A) If $-1 < x < 1$, then $f(x)$ satisfies

(p) $0 < f(x) < 1$

(B) If $1 < x < 2$, then $f(x)$ satisfies

(q) $f(x) < 0$

(C) If $3 < x < 5$, then $f(x)$ satisfies

(r) $f(x) > 0$

(D) If $x > 5$, then $f(x)$ satisfies

(s) $f(x) < 1$

Answer

	p	q	r	s
A	☒	☐	☒	☒
B	☐	☒	☐	☒
C	☐	☒	☐	☒
D	☒	☐	☒	☒

65. Let (x, y) be such that

$$\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}.$$

Match the statements in **Column I** with statements in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

(A) If $a = 1$ and $b = 0$, then (x, y)

(B) If $a = 1$ and $b = 1$, then (x, y)

(C) If $a = 1$ and $b = 2$, then (x, y)

(D) If $a = 2$ and $b = 2$, then (x, y)

Column II

(p) lies on the circle $x^2 + y^2 = 1$

(q) lies on $(x^2 - 1)(y^2 - 1) = 0$

(r) lies on $y = x$

(s) lies on $(4x^2 - 1)(y^2 - 1) = 0$

Answer

	p	q	r	s
A	☒	☐	☐	☐
B	☐	☒	☐	☐
C	☒	☐	☐	☐
D	☐	☐	☐	☒

66. Match the statements in **Column I** with the properties in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

Column II

- | | |
|--|----------------------------------|
| (A) Two intersecting circles | (p) have a common tangent |
| (B) Two mutually external circles | (q) have a common normal |
| (C) Two circles, one strictly inside the other | (r) do not have a common tangent |
| (D) Two branches of a hyperbola | (s) do not have a common normal |

Answer

	p	q	r	s
A	☒	☒	☐	☐
B	☒	☒	☐	☐
C	☐	☒	☒	☐
D	☐	☒	☒	☐